

ADVANCE

SIMULATED ADVANCE — aggregated procurement commitment for the incumbent extruders

The simulated Ministry-grade evidence shows incumbent extruders with spare capacity and an equivalent alloy/temper, while target-profile demand is fragmented across many projects; an aggregated, conditional offtake makes the die, tooling and qualification investment bankable without capital support.

SAU-H6-760429 · Demand aggregation, procurement commitment or conditional offtake (route 4) · Mode: Ministry simulation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Aluminium-alloy bars, rods and profiles, other than hollow profiles
HS H6 / 760429

Unchanged public conclusion
INVESTIGATE — binding constraint unresolved

Target-specification demand 6 kt	Effective qualified capacity 2.497 kt
Published capability distance D* 0.25 ratio	Minimum effective support S* 0 SAR m

Decision conditions
Aggregate committed project offtake of at least 3.6 kt across the named buyers before the incumbents invest in dies and qualification. [1 more; see the full record](#)

Kill conditions
Committed offtake falls below the die and qualification break-even volume. [1 more; see the full record](#)

Evidence passport ledger: [P-ALUPCO-760429](#) [P-COMTRADE-760429-M](#)
Snapshot: PUBLIC-SAU-H6-760429-2026-09-12 · As of: 2026-09-12

Supporting evidence pack

1. [Identity and specification boundary](#)
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Identity and specification boundary

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Unchanged public conclusion

INVESTIGATE — binding constraint unresolved

Brownfield priority to test

Test brownfield first; greenfield is not justified from public evidence.

Decision conditions

- No greenfield or financial support recommendation before effective capacity and target specification are resolved.

Kill conditions

- Demand is temporary, re-exported or below efficient scale
- Incumbent expansion already closes the specification-adjusted gap

Next evidence actions

- Saudi tariff-line and invoice description
- Importer/offtaker application and qualification requirements
- Producer grade, application-equivalence and local-availability matrix
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages

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- Delivered cost, import parity and unsupported Brownfield economics

Product identity

Source field	Value	Unit
HS revision	H6	
HS code	760429	
National tariff line	Unavailable Not established by the supplied evidence.	
Commercial name, English source	Aluminium-alloy bars, rods and profiles, other than hollow profiles	
Commercial name, Arabic	قضبان وعيدان وقطاعات من خلأط الألومنيوم، غير مجوفة	
Application boundary	HS 7604.29 identifies aluminium-alloy bars, rods and profiles other than the separately named hollow-profile subheading; alloy, geometry, temper, finishing, engineering application and customer liability remain unresolved.	
Target specification	6063/6061-T6 architectural and structural aluminium-alloy solid profiles to project drawings SYN-MINISTRY-ALU-PROFILES-001::target_specification	
Standard	SASO/ASTM B221 with EN 755 tolerances (synthetic design basis) SYN-MINISTRY-ALU-PROFILES-001::target_specification	
Alloy	6063 6061 SYN-MINISTRY-ALU-PROFILES-001::target_specification	
Temper	T6 SYN-MINISTRY-ALU-PROFILES-001::target_specification	
Application	curtain-wall, façade and structural framing systems SYN-MINISTRY-ALU-PROFILES-001::target_specification	
Zinc coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-ALU-PROFILES-001::target_specification	g/m ²

Source field	Value	Unit
Tin coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-ALU-PROFILES-001::target_specification	g/m ²
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-ALU-PROFILES-001::target_specification	mm
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-ALU-PROFILES-001::target_specification	µm
Width	Unavailable Not established by the supplied evidence. SYN-MINISTRY-ALU-PROFILES-001::target_specification	mm
Customer qualification required	Yes SYN-MINISTRY-ALU-PROFILES-001::target_specification	

Classification revision and concordance

Unavailable

Specification source spans

Unavailable

Demand and public trade signals

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Demand conclusion

Gross imports are a screening signal, not demand at the target specification.

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Source field	Value	Unit
Base demand	9.1 SYN-MINISTRY-ALU-PROFILES-001::demand	kt
Target-specification demand	6 SYN-MINISTRY-ALU-PROFILES-001::demand	kt
Downside demand	5.2 SYN-MINISTRY-ALU-PROFILES-001::demand	kt
Committed demand	3.6 SYN-MINISTRY-ALU-PROFILES-001::demand	kt
Announced demand	1.5 SYN-MINISTRY-ALU-PROFILES-001::demand	kt
Commitment probability	0.8 SYN-MINISTRY-ALU-PROFILES-001::demand	ratio
Observation year	2024	year
Gross import value	51.776086	USD m
Gross import quantity	9.189841	kt
Named buyers	Unavailable Not established by the supplied evidence.	
Buyer concentration	Unavailable Not established by the supplied evidence.	
Demand timing	Unavailable Not established by the supplied evidence.	
Year	Import value, USD m	Import quantity, kt
2022	33.566957	8.024333
2023	29.116347	5.098311
2024	51.776086	9.189841
Trade coverage and measurement		
Domestic-origin exports separated	No	
Execution ceiling	DEGRADED where continuity or tariff-line detail is required; 2021 (H5) excluded by the latest-revision H6 rule Partner detail: PARTNER_DETAIL_OBSERVED (P-WITS-760429-PARTNERS).	

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Trade-flow basis	gross
Missing years	None recorded; no completed review is implied.
Monthly partner/tariff-line detail available	No
Quantity is comparable	Yes
Re-exports separated	No

Production, retained flows and re-exports

Domestic-origin exports (kt)	Unavailable
Domestic production (kt)	Unavailable
Reference year	2024
Re-exports (kt)	Unavailable
Retained imports (kt)	Unavailable
Source evidence references	None recorded; no completed review is implied.

Supplier diagnostics

Partner concentration by value (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier value share	Not calculable
Partner concentration by quantity (HHI)	0.2956
Largest supplier quantity share	0.4787
Bulk unit-value range (USD/t)	Not calculable
Small-volume high-unit-value observation	
Trading partner	Japan
Quantity (kt)	Not calculable
Unit value (USD/t)	2042400

Partner detail (2024 imports)

Observed from wits_trade — 43 partner rows

Acquisition attempt references • None recorded; no completed review is implied.

Observed partner rows	43
Observed evidence passport	P-WITS-760429-PARTNERS
Partner snapshot identifier	PARTNERS-SAU-WITS-TRADE-2026-09-12
Source explanation	Unavailable
Source identifier	wits_trade
State	PARTNER_DETAIL_OBSERVED
Recorded unit	• 760429• imports• 2024

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Recorded basis Synthetic importer records for the target profiles within the public 2024 import quantity

Named buyers	
• Synthetic buyer identifier	SYN-BUYER-01
Quantity (kt)	0.9
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-02
Quantity (kt)	0.7
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-03
Quantity (kt)	0.6
Buyer segment	structural fabricator
• Synthetic buyer identifier	SYN-BUYER-04
Quantity (kt)	0.5

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Buyer segment	structural fabricator
• Synthetic buyer identifier	SYN-BUYER-05
Quantity (kt)	0.5
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-06
Quantity (kt)	0.4
Buyer segment	industrial framing
• Synthetic buyer identifier	SYN-BUYER-07
Quantity (kt)	0.4
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-08
Quantity (kt)	0.4
Buyer segment	structural fabricator
• Synthetic buyer identifier	SYN-BUYER-09
Quantity (kt)	0.3
Buyer segment	industrial framing
• Synthetic buyer identifier	SYN-BUYER-10
Quantity (kt)	0.3
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-11
Quantity (kt)	0.3
Buyer segment	structural fabricator
• Synthetic buyer identifier	SYN-BUYER-12
Quantity (kt)	0.2
Buyer segment	industrial framing
• Synthetic buyer identifier	SYN-BUYER-13

Quantity (kt)	0.2
Buyer segment	facade contractor
• Synthetic buyer identifier	SYN-BUYER-14
Quantity (kt)	0.2
Buyer segment	industrial framing
• Synthetic buyer identifier	SYN-BUYER-15
Quantity (kt)	0.1
Buyer segment	structural fabricator

Domestic supply and qualified capacity

Supply conclusion

Public producer observations; nameplate is not qualified capacity.

Named producer	Value
ALUPCO	Evidence class C
	Observed nameplate capacity, t/y Unavailable
	Nominal-capacity evidence reference Unavailable
	Nominal-capacity status unresolved
	Process family fabricated_aluminium
	Published process route aluminium extrusion
	Published standards Unavailable
	P-ALUPCO-760429
Al Taiseer Group TALCO	Evidence class C
	Observed nameplate capacity, t/y Unavailable
	Nominal-capacity evidence reference Unavailable
	Nominal-capacity status unresolved
	Process family fabricated_aluminium
	Published process route aluminium profile extrusion
	Published standards Unavailable
	P-TALCO-760429

Observed adjacency signals

Description	Domestic aluminium extrusion and profile production are publicly described.
Evidence references	◦ P-ALUPCO-760429◦ P-TALCO-760429
Signal type	core_process

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Source field	Value	Unit
Simulated nameplate capacity	40 SYN-MINISTRY-ALU-PROFILES-001::plant_line	kt
Availability factor	0.9 SYN-MINISTRY-ALU-PROFILES-001::plant_line	ratio
Yield factor	0.85 SYN-MINISTRY-ALU-PROFILES-001::plant_line	ratio
Qualification share	0.16 SYN-MINISTRY-ALU-PROFILES-001::plant_line	ratio
Market allocation share	0.51 SYN-MINISTRY-ALU-PROFILES-001::plant_line	ratio
Current utilisation	0.62 SYN-MINISTRY-ALU-PROFILES-001::plant_line	ratio
Description	Incumbent die, tooling and qualification for the aggregated profile order book; no new nameplate SYN-MINISTRY-ALU-PROFILES-001::upgrade	
Incremental capacity	3.6 SYN-MINISTRY-ALU-PROFILES-001::upgrade	kt
Upgrade schedule	9 SYN-MINISTRY-ALU-PROFILES-001::upgrade	months
Effective qualified capacity	2.497	kt
Qualified available supply	2.5	kt

Gap diagnosis and false-positive controls

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Quantity gap

Source field	Value	Unit
Specification-adjusted gap	3.5	kt

Source reference	Status	Value
EX-01_HETEROGENEOUS_RESIDUAL	Not satisfied	The commercial product cannot be separated from a highly heterogeneous residual code and no product-level evidence is available. SYN-MINISTRY-ALU-PROFILES-001::hard exclusion inputs
EX-02_MARKET_BELOW_MES	Not satisfied	The sustainable downside market is below minimum efficient scale and no credible export contract exists. SYN-MINISTRY-ALU-PROFILES-001::hard exclusion inputs
EX-03_UNSATISFIABLE_HARD_GATE	Not satisfied	A legal, safety, environmental, IP or customer-qualification hard gate cannot be satisfied. SYN-MINISTRY-ALU-PROFILES-001::hard exclusion inputs
EX-04_IDLE_EQUIVALENT_CAPACITY	Not satisfied	Domestic production meets the required specification with material idle capacity and no binding market failure is evidenced. SYN-MINISTRY-ALU-PROFILES-001::hard exclusion inputs

Source reference	Status	Value
EX-05_TRANSITORY_OR_MEASUREMENT	Not satisfied	The apparent gap is dominated by re-export, one-off project demand, temporary price arbitrage or classification discontinuity. SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs
EX-06_REDUNDANCY_OR_CROWD_OUT	Not satisfied	The intervention would create unacceptable redundant capacity or crowd out a more efficient incumbent. SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs

Test the apparent gap

Source reference	Status	Value
hard_exclusion	Not satisfied	HARD_EXCLUSION_NOT_SATISFIED SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs
false_or_measurement_gap	Not satisfied	FALSE_OR_MEASUREMENT_NOT_ESTABLISHED SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs
equivalent_idle_qualified_supply	Not satisfied	EQUIVALENT_QUALIFIED_SUPPLY_NOT_ESTABLISHED SYN-MINISTRY-ALU-PROFILES-001::demand SYN-MINISTRY-ALU-PROFILES-001::equivalence
uneconomic_at_efficient_scale	Not satisfied	ECONOMIC_ROUTE_EXISTS SYN-MINISTRY-ALU-PROFILES-001::route_evidence
structural_overcapacity	Not satisfied	STRUCTURAL_OVERCAPACITY_NOT_ESTABLISHED SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs

Source reference	Status	Value
generic_capacity_contradicted	Not satisfied	GENERIC_CAPACITY_NOT_CONTRADICTED P-COMTRADE-760429-X

Capability and route alternatives

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Source field	Value	Unit
Known coverage K	1	ratio
Unknown weight U	0	ratio
Uncertainty penalty (lambda)	0.5	ratio
Minimum known coverage	0.7	ratio
Published capability distance D*	0.25	ratio

Source field	State	Weight
Feedstock chemistry	0	0.1
Core process route	0	0.15
Equipment operating envelope	1	0.15
Finishing and specification control	1	0.15
Quality, laboratories and metrology	1	0.1
Certification and customer qualification	2	0.15
Capacity and timing	0	0.1

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Source field	State	Weight
Utilities, environment and permitting	0	0.05
Skills and market integration	1	0.05

Sector-profile hard gates

Alloy	RESOLVED
Forming and fabrication	RESOLVED
Heat treatment	RESOLVED
Joining and finishing	RESOLVED
Engineering certification	RESOLVED
Customer liability	RESOLVED

Unresolved hard gates • None recorded; no completed review is implied.

Known hard-gate failures • None recorded; no completed review is implied.

No intervention

Fails

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Fails
Additionality	Passes
Policy permissibility	Passes

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

passes

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- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Information and market linkage

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
-

Certification, testing and quality

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Demand aggregation and offtake

Passes

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes
Unsupported NPV	1.19
Unsupported IRR	0.16319
Minimum effective support S*	0

Lower-route precedence: No lower route recorded

Incremental national value	31
Capacity ratio	1.1731
Warning threshold	1.25
Passes the recorded test	Yes

[SYN-MINISTRY-ALU-PROFILES-001::route_evidence](#)

Brownfield upgrade or expansion

Fails

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Fails
Unsupported NPV	1.19
Unsupported IRR	0.16319

Minimum effective support S* 0

Lower-route precedence: Demand aggregation and offtake

Incremental national value 31

Capacity ratio 1.1725

Warning threshold 1.25

Passes the recorded test Yes

- The route is prohibited, disproportionate or unacceptably distortive.
- A lower-intervention route fully resolves the binding constraint.

[SYN-MINISTRY-ALU-PROFILES-001::route_evidence](#)

Technology partnership or joint venture

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: Demand aggregation and offtake

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Greenfield capacity

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: Demand aggregation and offtake

Incremental national value Not calculable

Not calculable

- Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.
- A lower-intervention route fully resolves the binding constraint.

Shared enabling infrastructure

Fails

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Fails
Additionality	Passes
Policy permissibility	Passes
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: Demand aggregation and offtake

Incremental national value	35.28
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Capacity ratio	1.074
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Warning threshold	1.25
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Passes the recorded test	Yes
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- The route passes its gates but does not fully remove the binding constraint; it does not block higher routes.
- A lower-intervention route fully resolves the binding constraint.

[SYN-MINISTRY-ALU-PROFILES-001::shared_enabler](#)

External dependency already evaluated by the engine; not a local passport.

- SYN-MINISTRY-ALU-FOIL-001::shared_enabler

Public route comparison

No intervention

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
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Unsupported IRR	Not calculable
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Minimum effective support S*	Not calculable
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Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
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Not calculable

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- Administrative, classification or regulatory barrier evidence is required.

Information and market linkage

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified information, linkage or matching constraint is required.

Certification, testing and quality

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified specification, certification, testing or qualification constraint is required.

Demand aggregation and offtake

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable
Not calculable	
A verified demand-fragmentation or offtake constraint is required.	

Brownfield upgrade or expansion

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
Not calculable	
- Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required. - Downside route economics are unavailable.	

Technology partnership or joint venture

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified technology constraint and major-line/JV adjacency are required.

Greenfield capacity

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
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Not calculable

- Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
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Unsupported IRR	Not calculable
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Minimum effective support S*	Not calculable
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Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
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Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Counterfactual

Can the incumbent meet the specification without capital investment?	No
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Does downside demand support both incumbent and entrant?	No
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Does entry displace efficient domestic production?	Yes
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Is a technology partnership superior to expansion or greenfield?	No
Missing capabilities	• equipment_envelope • finishing_spec_control • qa_lab_metrology • certification_customer_qualification • skills_market_integration
Brownfield versus greenfield comparison	
Incremental capacity	3.6
Upgrade schedule	9
Unsupported NPV	1.19
Minimum effective support S*	0
Greenfield alternative	Unavailable

Economics and intervention

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Source field	Value	Unit
Unsupported NPV	1.19	SAR m
Unsupported IRR	0.16319	ratio
Minimum effective support S*	0	SAR m
Supported NPV	1.19	SAR m
Supported IRR	0.16319	ratio
Hurdle rate	0.12	ratio
Separate downside result	Not calculable Cannot be calculated from the supplied evidence.	SAR m

National-value comparison

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National benefits	39
National costs	8
Incremental national value	31
Positive result	Yes
Components	
Displacement	2
Domestic value added	22
Exports	4
Fiscal receipts	4
Government cost	1
Knowledge and skills	3
Resilience value	6
Resources and environment	1
Risk allowance	4

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SIMULATED — NOT MINISTRY EVIDENCE

Cash flows without support (SAR m), in source order • -9.5• 2.6• 2.6• 2.6• 2.6• 2.6• 2.6

Currency SAR_m

Hurdle rate 0.12

Minimum efficient scale (kt) 3

National value

Displacement 2

Domestic value added 22

Exports 4

Fiscal receipts 4

Government cost 1

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Knowledge and skills	3
Resilience value	6
Resources and environment	1
Risk allowance	4
Support instrument	none — procurement coordination only

[SYN-MINISTRY-ALU-PROFILES-001::economics](#)

Cash-flow inputs in source order

• Route code	4
Downside cash flows (SAR m), in source order	◦ -9.5◦ 2.6◦ 2.6◦ 2.6◦ 2.6◦ 2.6◦ 2.6
Hurdle rate	0.12
Recorded basis	Aggregated, conditional project offtake enabling incumbent die, tooling and qualification investment without financial support.
• Route code	5
Recorded basis	Capital support for expansion is disproportionate while incumbents hold spare extrusion capacity.

[SYN-MINISTRY-ALU-PROFILES-001::route_evidence](#)

Competition and public policy

Class D simulation · SYN-MINISTRY-ALU-PROFILES-001

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Source field	Value	Unit
Capacity to downside demand ratio	1.1725	ratio
Displacement cost	2	SAR m

Source field	Value	Unit
Sunset date	Unavailable Not established by the supplied evidence.	
Capacity to downside demand ratio	1.1725	
Warning threshold	1.25	
Warning is triggered	No	
Triggers the default warning	Yes	
Displacement cost	2	
Non-additionality assessment		
Status	passes	
Recorded basis	route_record_additionality	
Concentration before and after		
Concentration before entry (HHI)	0.31	
Concentration after entry (HHI)	0.29	
Change	-0.02	
Status	REPORTED	
Source note	Lower concentration is not equated with benefit.	

Calculation and recommendation do not authorize public support.

Complete rule execution ledger

R0 · Identity gate

Full · Fires

HS revision H6 and HS6 760429 are frozen in the snapshot; specification/application remains separately gated.

Continue to screening while preserving the unresolved decision object.

R1-F · Persistent retained exposure — full

Full · Does not fire

Full persistence test did not fire.

Do not claim the full persistence rule.

Missing years • None recorded; no completed review is implied.

R1-D · Persistence signal — degraded

Degraded · Fires

3 positive observed years within the configured window; no interpolation used.

Generate INVESTIGATE only; confidence capped at C.

Positive observed years • 2022 • 2023 • 2024

Confidence ceiling C

R2 · Quantity-led expansion

Full · Fires

Quantity-led expansion signal fires.

Separate structural volume growth from price movement.

First year 2023

Final year 2024

Observed period (years) 1

Change in log value 0.5756

Change in log quantity 0.5892

Change in log unit value -0.0136

Quantity contribution share 0.9775

Quantity compound annual growth rate 0.8025

R3 · Supplier concentration

Full · Fires

External supply is concentrated on the quantity basis; value concentration is NOT_CALCULABLE.

Generate a resilience/diversification review, not an automatic localisation recommendation.

Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Concentration threshold (HHI)	0.25
Largest-supplier threshold	0.5
Value	
Recorded basis	value
Status	Not calculable
Observation year	2024
Trade-flow basis	gross
Source	partner_observations
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	1
Eligible partner count	43
Source explanation	Eligible partner rows do not reconcile to the matching world aggregate value basis.
Quantity	
Recorded basis	quantity
Status	FULL
Observation year	2024
Trade-flow basis	gross

Source	partner_observations
Concentration index (HHI)	0.2956
Largest supplier share	0.4787
Top-two supplier share	0.6905
Valid observation coverage	1
Eligible partner count	43
Source explanation	Unavailable

R4-F · Product-tier heterogeneity — full

Disabled · Not evaluable

Partner-month × tariff-line cells are unavailable.

Do not fit clusters or claim grades.

R4-D · Unit-value dispersion — degraded

Degraded · Fires

Comparable annual partner unit values show descriptive dispersion; no cluster or grade conclusion.

Open specification research only; no grade conclusion.

Status	CALCULABLE
Calculation basis	partner_observations
Valid value coverage	1
Valid quantity coverage	1
Minimum valid-value coverage	0.7
Weighted first quartile (USD/t)	3426.3684
Weighted median (USD/t)	3426.3684
Weighted third quartile (USD/t)	6143.4362
Interquartile range (USD/t)	2717.0678
Bulk unit-value band (USD/t)	Not calculable

Potential outlier	
Trading partner	Japan
Unit value (USD/t)	2042400
Quantity share	0
Confirmed outlier	No
Source explanation	Unavailable

R5 · Domestic supply plus continued imports

Degraded · Fires

Verified domestic capability coexists with material gross imports.

Test specification, qualification, capacity, price, application and allocation mismatch.

Retained imports (kt)	Not calculable
Net import exposure (kt)	Not calculable
Apparent consumption (kt)	Not calculable
Retained imports / apparent consumption	Not calculable
Calculation basis	Not calculable
Unavailable inputs	- domestic_origin_exports_kt • domestic_production_kt - reexports_kt • retained_imports_kt
Reasons for unavailability	
Retained imports (kt)	reexports_kt • retained_imports_kt
Net import exposure (kt)	domestic_origin_exports_kt • retained_imports_kt
Apparent consumption (kt)	- domestic_origin_exports_kt • domestic_production_kt - retained_imports_kt
Retained imports / apparent consumption	apparent_consumption_kt • retained_imports_kt
Configured threshold	0.2

Source explanation	Domestic production quantity and retained-import flow are absent from the frozen public snapshot; gross imports cannot establish apparent consumption.
--------------------	--

R6 · Capacity pressure

Disabled · Not evaluable

Effective utilisation and target-specification shortage are not public.

Obtain line-level availability, yield, qualification share and allocation.

R7 · Latent domestic capacity

Disabled · Not evaluable

Idle qualified capacity and specification equivalence are not established publicly.

Do not assume latent capacity.

R8 · Committed future demand

Disabled · Not evaluable

Awarded/financed demand at the target specification is unavailable.

Keep base, committed and announced demand separate when provided.

R9-S · Coarse incumbent adjacency screen

Full · Fires

A verified Saudi plant is in the same process family with additional capability signals.

Open the full line-level capability assessment; do not publish D* from screening alone.

Same process family	Yes
Qualifying signal count	1
Known hard-gate failure	No

R10 · Strategic criticality

Degraded · Fires

A resilience review is warranted by concentration; formal criticality still requires authority confirmation.

Keep commercial viability and strategic value separate.

Criticality status	Not calculable
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Criticality evidence reference	Unavailable
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R11 · Economic exclusion / generic-capacity warning

Degraded · Does not fire

Export/import ratio or established nameplate capability is NOT_CALCULABLE; no generic-capacity exclusion fires.

Reject generic support or move only a named specialty exception to investigation.

Export/import value ratio	1.5579
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Export/import ratio threshold	50
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Calculated export/import value ratio	1.5579
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Disclosed export/import value ratio	Unavailable
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Disclosed ratio is consistent	Unavailable
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Ratio basis	gross_trade_value
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Ratio status	CALCULABLE
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Ratio explanation	Unavailable
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Established nominal capacity

Established by evidence	No
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Total observed nominal capacity (t/year)	Not calculable
--	----------------

Named producers	• None recorded; no completed review is implied.
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Evidence references	• None recorded; no completed review is implied.
---------------------	--

Source explanation	No positive observed A/B/C producer nameplate exists.
--------------------	---

R12 · Evidence-value trigger

Degraded · Fires

5 named facts could change the route.

Prioritise the smallest evidence request with a plausible route effect; quantify EVSI when cost and value inputs exist.

Named missing facts

- Saudi tariff-line and invoice description
- Importer/offtaker application and qualification requirements
- Producer grade, application-equivalence and local-availability matrix
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages
- Delivered cost, import parity and unsupported Brownfield economics

Evidence needed

- Saudi tariff-line and invoice description

Evidence-need identifier identity/tariff-line

Blocked decision field product_identity

Effect on route Can resolve the decision object and change the admissible route.

Evidence references ◦ P-WCO-760429

Numerical value of information Not calculable

- Importer/offtaker application and qualification requirements

Evidence-need identifier target specification/application

Blocked decision field demand_at_required_specification

Effect on route Can establish target demand and change state or route.

Evidence references ◦ None recorded; no completed review is implied.

Numerical value of information Not calculable

- Producer grade, application-equivalence and local-availability matrix

Evidence-need identifier line-level production or producer-grade matrix

Blocked decision field	domestic_supply_or_capability
Effect on route	Can establish domestic equivalence and change the incumbent route.
Evidence references	◦ P-ALUPCO-760429◦ P-TALCO-760429
Numerical value of information	Not calculable
• Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages	
Evidence-need identifier	capacity/availability/allocation
Blocked decision field	domestic_supply_or_capability
Effect on route	Can distinguish no action, brownfield expansion, and new capacity.
Evidence references	◦ None recorded; no completed review is implied.
Numerical value of information	Not calculable
• Delivered cost, import parity and unsupported Brownfield economics	
Evidence-need identifier	route economics
Blocked decision field	route_economics
Effect on route	Can determine whether the plausible route proceeds unsupported.
Evidence references	◦ None recorded; no completed review is implied.
Numerical value of information	Not calculable

Simulated R6–R8 ledger

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

R5 · Retained import penetration

Full · Fires

Synthetic retained-import penetration meets the configured R5 threshold.

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SIMULATED — NOT MINISTRY EVIDENCE

Use only when production and retained-flow blocks are declared.

Retained imports (kt)	8.6
Domestic production (kt)	38
Gross import quantity	9.189841
Retained-import penetration	0.3359375
Apparent consumption (kt)	25.6
Minimum retained-import penetration	0.2

R6 · Capacity pressure

Degraded · Does not fire

Synthetic R6 utilisation/shortage proxy does not meet both configured thresholds; sustained period is NOT_CALCULABLE.

Use as a DEGRADED capacity-pressure signal only; it cannot by itself support ADVANCE without a sustained period.

Effective utilisation	0.62
Minimum effective utilisation	0.85
Target-specification demand	6
Effective qualified capacity	2.497
Effective-capacity source field	effective_qualified_capacity_kt
Shortage ratio	1.4029
Shortage denominator	effective_qualified_capacity_kt
Minimum specification-matched shortage	0.1
Sustained period	Not calculable

R7 · Latent domestic capacity

Full · Fires

Synthetic utilisation and specification equivalence meet the configured latent-capacity test.

Test no-support, market linkage, procurement, or barrier removal only when equivalence and latent capacity are established.

Effective utilisation	0.62
Maximum effective utilisation	0.7
Specification equivalence	Yes
Qualified available supply	2.5

R8 · Committed future demand

Full · Fires

Committed and announced layers are disclosed, but base demand, commitment probability, and minimum efficient scale govern R8 calculability.

Do not infer probability-adjusted demand addition or MES fill; keep committed and announced demand separate.

Base demand	9.1
Committed demand	3.6
Announced demand	1.5
Target-specification demand	6
Downside demand	5.2
Commitment probability	0.8
Probability-adjusted committed demand (kt)	2.88
Probability-adjusted demand addition	0.316484
Minimum probability-adjusted demand addition	0.2
Minimum efficient scale (kt)	3
Minimum efficient scale coverage	0.96
Minimum scale-coverage threshold	0.25

Evidence passports and contradictions

6 public records; 19 synthetic records.

Contradiction register

Public evidence contradictions

No public contradictions recorded.

Synthetic evidence contradictions

No synthetic contradictions are recorded for this simulation.

P-ALUPCO-760429

About ALUPCO

Source field	Value
Source	producer_alupco
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Verbatim span from DocumentRecord DOC-PRODUCER-ALUPCO-135870d0e4c7-0261d54ca8a6.
Supporting claims	DOMESTIC_PRODUCT_PORTFOLIO
Contradiction register	Unavailable
Source reference	https://alupco.com/about/

P-COMTRADE-760429-M

UN Comtrade Saudi imports rows for HS 760429

Source field	Value
Source	UN Comtrade

Source field	Value
Class	B
Status	calculated
Source period	2022/2023/2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt.
Supporting claims	• TRADE_QUANTITY• TRADE_VALUE
Contradiction register	Unavailable
Source reference	https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&period=2024&flowCode=M&cmdCode=AG6&partnerCode=0&motCode=0&customsCode=C00

P-COMTRADE-760429-X

UN Comtrade Saudi exports rows for HS 760429

Source field	Value
Source	UN Comtrade
Class	B
Status	calculated
Source period	2022/2023/2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt.
Supporting claims	• EXPORT_IMPORT_RATIO• TRADE_QUANTITY• TRADE_VALUE
Contradiction register	Unavailable
Source reference	https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&period=2024&flowCode=X&cmdCode=AG6&partnerCode=0&motCode=0&customsCode=C00

P-TALCO-760429

TALCO in brief

Source field	Value
Source	producer_altaiseer_talco
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Verbatim span from DocumentRecord DOC-PRODUCER-ALTAISEER-TALCO-b68e0eef2440-b339b1ac8d87.
Supporting claims	DOMESTIC_PRODUCT_PORTFOLIO
Contradiction register	Unavailable
Source reference	https://altaiseer.com/en/%d8%aa%d8%a7%d9%84%d9%83%d9%88-%d9%81%d9%8a-%d8%b3%d8%b7%d9%88%d8%b1/

P-WCO-760429

HS Nomenclature 2022 — Chapter 76

Source field	Value
Source	wco_hs_nomenclature
Class	B
Status	observed
Source period	2022
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Verbatim span from DocumentRecord DOC-WCO-HS-NOMENCLATURE-69bd7323c7aa-bd350605a78a.
Supporting claims	TARGET_PRODUCT_IDENTITY
Contradiction register	Unavailable

Source field	Value
Source reference	https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/nomenclature/instruments-and-tools/hs-nomenclature-2022/2022/1576_2022e.pdf?la=en

P-WITS-760429-PARTNERS

WITS Saudi 2024 import partner rows for HS 760429

Source field	Value
Source	World Bank WITS
Class	B
Status	calculated
Source period	2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	World aggregate excluded; value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt. Partner detail state: PARTNER_DETAIL_OBSERVED (wits_trade; 43 partner rows).
Supporting claims	SUPPLIER_CONCENTRATION
Contradiction register	Unavailable
Source reference	https://wits.worldbank.org/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/760429

SYN-MINISTRY-ALU-PROFILES-001::buyer_allocation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Buyer Allocation

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
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Source field	Value
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: buyer_allocation
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::capability_states

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Capability States

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: capability_states
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::class_if_confirmed

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Class If Confirmed

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: class_if_confirmed
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::competition_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Competition Inputs

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: competition_inputs
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::counterfactual

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Counterfactual

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: counterfactual
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::decision_specific_hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision Specific Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
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Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: decision_specific_hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::demand

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Demand

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: demand
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::economics

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Economics

Source field	Value
Source	DEMO_GENERATOR

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: economics
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::equivalence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Equivalence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: equivalence
Contradiction register	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Evsi

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: evsi
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::hard_exclusion_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Exclusion Inputs

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: hard_exclusion_inputs
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::plant_line

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Plant Line

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
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Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: plant_line
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::production_and_retained_flows

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Production And Retained Flows

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: production_and_retained_flows
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::route_evidence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Route Evidence

Source field	Value
Source	DEMO_GENERATOR

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: route_evidence
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::shared_enabler

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Shared Enabler

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: shared_enabler
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::target_specification

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Target Specification

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: target_specification
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::tariff_line_allocation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Tariff Line Allocation

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: tariff_line_allocation
Contradiction register	Unavailable

SYN-MINISTRY-ALU-PROFILES-001::upgrade

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Upgrade

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: upgrade
Contradiction register	Unavailable

External dependency already evaluated by the engine; not a local passport. SYN-MINISTRY-ALU-FOIL-001::shared_enabler

Conditions, kill conditions and next facts

Class D simulation · SYN-MINISTRY-ALU-PROFILES-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision conditions

- Aggregate committed project offtake of at least 3.6 kt across the named buyers before the incumbents invest in dies and qualification.
- No financial support is granted; the intervention is procurement coordination only.

Kill conditions

- Committed offtake falls below the die and qualification break-even volume.
- Buyers revert to imported profiles after qualification.

Next evidence actions

- None

Source field	Value	Unit
Approximate EVSI	11.45	SAR m
Route-change probability	0.45	
Value difference (SAR m)	31	
Evidence acquisition cost (SAR m)	1	
Delay cost (SAR m)	1.5	
Approximate EVSI	11.45	
Positive result	Yes	
Next fact to establish	Confirm project-level profile specifications and the incumbents' qualified allocation	

Decision history and expert overrides

None recorded; no completed review is implied.

Authority and reproducibility

Authority and versions

Source reference	Value
Methodology	docs/authority/Industrial_Opportunity_Resolution_Methodology_Final_KSA.docx 5717cbd42acc9947ce5e450013719275acb7ed1470847b21fb2cc547c8ac4ce9
Project	0.2.0
Thresholds	1.3.0
Sector profiles	1.1.0
Evidence policy	1.4.0
UI strings	1.7.0

Evidence boundary

Public decision uses public evidence only	Yes
Synthetic branch isolation	Yes
Scenario reconciliation	
Status	PASS
Scenario identifier	SYN-MINISTRY-ALU-PROFILES-001
Opportunity identifier	SAU-H6-760429
Public snapshot identifier	PUBLIC-SAU-H6-760429-2026-09-12
Recorded checks	

• Rule identifier	target_spec_demand_within_public_imports
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	target_spec_demand_kt <= latest_public_imports_kt
Recorded result	PASS
Blocking check	Yes

Source inputs	
Target-specification demand	6
Latest public observation year	2024
Latest public imports (kt)	9.189841
Source explanation	6 kt <= 9.18984 kt.
• Rule identifier	line_nameplate_within_disclosed_public_capacity
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	plant_line.nameplate_kt <= sum(non_null installed_capacity_tpy) / 1000
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Line nominal capacity (kt)	40
Disclosed installed capacity (t/year)	◦ Unavailable◦ Unavailable
Publicly disclosed nominal capacity (kt)	Unavailable
Expansion assumption applied	No
Source explanation	No numeric public installed nameplate is disclosed.
• Rule identifier	capacity_factors_within_unit_interval
Source	Core 06 §10 item 3 lines 226-233; Core 06 §6.2 lines 148-156; Core 05 §8 lines 197-203
Recorded formula	for every capacity factor: 0 <= factor <= 1
Recorded result	PASS
Blocking check	Yes
Source inputs	
Availability factor	0.9
Yield factor	0.85

Qualification share	0.16
Market allocation share	0.51
Current utilisation	0.62
Source explanation	All declared capacity factors are within [0,1].
• Rule identifier	qualified_availability_within_physical_output
Source	Core 06 §5.1 lines 92-101; Core 06 §10 item 3 lines 226-233
Recorded formula	qualified_available_kt <= nameplate_kt * availability * yield
Recorded result	PASS
Blocking check	Yes
Source inputs	
Qualified available supply	2.5
Simulated nameplate capacity	40
Availability factor	0.9
Yield factor	0.85
Physical output ceiling (kt)	30.6
Source explanation	2.5 kt <= 30.6 kt.
• Rule identifier	demand_layers_remain_separate
Source	Core 04 §2.6 lines 102-112
Recorded formula	observe downside_demand_kt <= target_spec_demand_kt and committed_demand_kt <= target_spec_demand_kt; no decision effect
Recorded result	INFORMATIONAL
Blocking check	No
Source inputs	
Target-specification demand	6
Downside demand	5.2
Committed demand	3.6

Base demand	9.1
Downside demand within target demand	Yes
Committed demand within target demand	Yes
Source explanation	Demand layers are reported separately; no ordering constraint is enforced.
• Rule identifier	tariff_line_allocation_sums_to_public_hs6_total
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	sum(tariff_line_allocation.lines.quantity_kt) == latest public imports_kt
Recorded result	PASS
Blocking check	Yes
Source inputs	
Total quantity (kt)	9.189841
Latest public imports (kt)	9.189841
Total value (USD m)	Unavailable
Latest public import value (USD m)	51.776086
Source explanation	Tariff-line quantities sum to 9.18984 kt against public imports 9.18984 kt.
• Rule identifier	buyer_allocation_within_public_imports
Source	Core 06 §5.1 lines 92-101
Recorded formula	sum(buyer_allocation.buyers.quantity_kt) <= latest public imports_kt
Recorded result	PASS
Blocking check	Yes
Source inputs	
Total quantity (kt)	6

Latest public imports (kt)	9.189841
Source explanation	Buyer quantities sum to 6 kt within public imports 9.18984 kt.
• Rule identifier	expansion_assumption_disclosed_and_bounded
Source	Core 06 §5.1 lines 92-101; KL-38
Recorded formula	planned_nameplate_kt >= disclosed public nameplate; commissioning_year >= public as_of year; plant_line.nameplate_kt <= planned_nameplate_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No expansion_assumption block is declared.
• Rule identifier	retained_flows_reconcile_to_public_trade
Source	Core 06 §5.1; methodology §3.3
Recorded formula	retained_imports_kt <= imports_kt; retained + reexports == imports when reexports numeric; domestic_origin_exports_kt <= public exports_kt
Recorded result	PASS
Blocking check	Yes
Source inputs	
Retained imports (kt)	8.6
Re-exports (kt)	0.589841
Domestic-origin exports (kt)	21
Latest public imports (kt)	9.189841
Latest public exports (kt)	21.318251
Source explanation	Retained flows reconcile to public trade marginals.
• Rule identifier	base_demand_and_commitment_probability_valid
Source	Core 06 §5.2; Core 07 §3 R8

Recorded formula	0 <= commitment_probability <= 1; base_demand_kt > 0; minimum_efficient_scale_kt > 0 when declared
Recorded result	PASS
Blocking check	Yes
Source inputs	
Base demand	9.1
Commitment probability	0.8
Minimum efficient scale (kt)	3
Source explanation	Base demand, commitment probability and MES inputs are valid.
• Rule identifier	shared_enabler_declaration_valid
Source	Core 06 scenario contract 2.1.0 and methodology §8.3
Recorded formula	explicit route valuation and complete Class-D declaration
Recorded result	PASS
Blocking check	Yes
Source inputs	
Shared-enabler identifier	ENABLER-SYN-ALU-CASTHOUSE-001
Valuation route code	4
Dependent incremental national value (SAR m)	31
Source explanation	Shared-enabler declaration and explicit valuation source are valid.

Latest imports (USD million)	51.776086
Latest imports (kt)	9.189841
Public decision unchanged after simulation	Yes
Scenario expected-result comparison	
Expected result	
State	ADVANCE
Route code	4
Actual result	
State	ADVANCE
Route code	4
Results match	Yes